Thirumalasetty Karthik

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Summary

Final-year B.Tech CSE student building GenAI applications end to end: hybrid RAG with citation validation, tool-calling agents with regression evaluation, and MCP-based multi-agent orchestration, backed by measurable evaluation metrics.

Experience

AI Engineering Intern Jun. 2026 – Jul. 2026
RV University – Bengaluru | LLM-Based Digital Forensics | Supervisor: Dr. Manish Kumar *Remote*

- Architected an air-gapped **hybrid RAG** pipeline over **30,265** parsed forensic records (EVTX, PCAP, CSV), fusing Qdrant dense search and BM25 with IOC token boosting via **RRF**; **0.38–1.1s** retrieval-only latency on CPU.
- Engineered a citation-enforced Q&A layer on local **Llama-3-8B** (Ollama) with a two-stage relevance gate, prompt-injection isolation, post-generation **citation validation**, and a similarity-based confidence cap.
- Built a Streamlit investigator UI with background-threaded queries, multi-turn sessions, and automated Markdown/PDF reporting from a deduplicated evidence inventory, backed by automated test suites.

Projects

PolicyPilot – Insurance Q&A System *(Python, LangChain, ChromaDB, Groq, Gradio)* GitHub

- Built a multi-stage **RAG** pipeline: dual-query retrieval (original + LLM-rewritten) reranked locally with a **cross-encoder** (`ms-marco-MiniLM-L-6-v2`); added an admin dashboard with PDF and Markdown ingestion.
- Built an **LLM-as-judge** and keyword-based retrieval harness over **150** questions in 7 categories: **MRR 0.87**, **nDCG 0.85**, **91.5%** keyword coverage, **4.31/5** judged accuracy, with per-category scores to find the weakest question types.

EffectSentinel – Agent Execution Safety Gateway *(LangGraph, SQLite, FastAPI, Groq)* GitHub

- Architected a **25-scenario** golden evaluation harness and **EffectGuard** safety gateway for a **LangGraph ReAct** agent, enforcing **Policy-as-Code** preconditions, a frozen temporal benchmark, and state-level **SQLite** validation.
- Built a chaos fault injector (504 timeouts) and **SHA-256 idempotency** engine with verify-before-retry and WAL journaling, eliminating duplicate tickets and reducing financial loss from **125+ USD** (naive baseline) to **zero**.
- Developed a **FastAPI** dashboard streaming live **SSE** telemetry across 6 gauges, featuring claim-vs-ledger audit diffs, an automated **C0–C4** comparative benchmark matrix, and CSV export.

Autonomous Multi-Agent Trading Floor *(Python, OpenAI Agents SDK, MCP, FastAPI)* GitHub

- Built a 4-agent trading simulation on the **OpenAI Agents SDK**: concurrent persona-driven traders, each with a researcher sub-agent and its own LLM (DeepSeek V3, Gemini 2.5 Flash, Llama 3.3 70B, GPT-OSS 20B).
- Connected agents to **MCP** servers over stdio (accounts, market data, push alerts, web search, page fetch, LibSQL memory); built a FastAPI API and Vite/TypeScript dashboard alongside Gradio.

Technical Skills

Languages & Core: Python, C/C++, SQL, Git
LLMs & GenAI: RAG, Tool-Calling Agents, LangGraph, Multi-Agent Systems, MCP, OpenAI Agents SDK, Prompt Engineering, LLM Fine-Tuning (QLoRA), LangChain, PyTorch, Hugging Face (Transformers, PEFT)
Retrieval & Evaluation: Vector Databases (ChromaDB, Qdrant), BM25, Hybrid Retrieval (Dense+Sparse+RRF), Cross-Encoder Reranking, DeepEval, LLM-as-Judge Evaluation (MRR, nDCG)
Deployment & Tools: Docker, FastAPI, Flask, Streamlit, Gradio, REST APIs, SSE, OpenAI, Groq, Modal, Ollama
AI Development Tools: Claude Code, Cursor, Antigravity

Education

Indian Institute of Information Technology (IIIT), Manipur Aug. 2023 – June 2027
B.Tech in Computer Science and Engineering *Imphal, Manipur*

Jawahar Navodaya Vidyalaya (JNV), Krishnapuram Aug. 2016 – May 2023
Matriculation and Intermediate (MPC) *Nellore, AP*

Achievements & Leadership

- Qualified **Graduate Aptitude Test in Engineering (GATE) 2026** in Computer Science and Engineering.
- **PR & Outreach Lead, AHOUBA Techfest** – led PR and social media for IIITM's flagship 3-day tech fest, coordinating with colleges across Manipur to drive participation.
- Founded **TrueClip** (live site), an advertising agency that delivered 650+ UGC video assets across languages and, driving lead generation for brands expanding into tier-2/tier-3 markets.